CFA Level I Mock Exam 7.2 - Mark Scheme

Each question lists the correct choice and the original feedback for all three choices. Source excerpts accompany explanations where equations are PDF artwork.

Question 1

When a company has significant intangibles, asset-based valuation is most likely:

Correct answer: B

A. used to determine the value of off the books assets. - Incorrect

Incorrect Answer Feedback: Incorrect because an asset-based valuation of a company uses estimates of the market or fair value of the company's assets and liabilities. Thus, asset-based valuations work well for companies that do not have a high proportion of intangible or "off the books" assets.

B. less preferred than forward-looking cash flow valuation. - Correct

Correct Answer Feedback: Correct because when a company has significant intangibles, the analyst should prefer a forward-looking cash flow valuation.

C. appropriate to calculate a maximum value for the company. - Incorrect

Incorrect Answer Feedback: Incorrect because asset-based valuation may not consider some intangibles, thus it can give a "floor" value, not a maximum value, for a situation involving a significant amount of intangibles.

Question 2

The lead underwriter in an IPO most likely:

Correct answer: A

A. acts as the book builder for the offering. - Correct

Correct Answer Feedback: Correct because the lead underwriter is the book builder for the offering. Book building is a process where the investment bank then lines up subscribers who will buy the security. For large issues, a syndicate of investment banks and broker-dealers helps the lead underwriter build the book.

B. is not responsible for making a market in the securities. - Incorrect

Incorrect Answer Feedback: Incorrect because in the case of an IPO, the underwriter usually also promises to make a market in the security for about a month to ensure that the secondary market will be liquid and to provide price support, if necessary.

C. is prohibited from providing an opinion about the issue to their clients. - Incorrect

Incorrect Answer Feedback: Incorrect because investment banks often support their book building by providing investment information and opinion about the issuer to their clients and to the public.

Question 3

The value of a European put option is directly related to the:

Correct answer: C

A. risk-free interest rate. - Incorrect

Incorrect Answer Feedback: Incorrect because the value of a European put is inversely related to the risk-free interest rate.

B. value of the underlying. - Incorrect

Incorrect Answer Feedback: Incorrect because the value of a European put option is inversely related to the value of the underlying.

C. volatility of the underlying. - Correct

Correct Answer Feedback: Correct because the value of a European put is directly related to the volatility of the underlying.

Question 4

If it is difficult to find a buyer or a seller for an asset, the asset most likely trades on:

Correct answer: A

A. brokered markets. - Correct

Correct Answer Feedback: Correct because brokered markets are markets in which brokers arrange trades among their clients. Brokers organize markets for instruments for which finding a buyer or seller willing to trade is difficult because the instruments are unique.

B. order-driven markets. - Incorrect

Incorrect Answer Feedback: Incorrect because organizing order-driven markets for these instruments (assets for which finding a buyer or seller is difficult) is not sensible because too few traders would submit orders for them.

C. quote-driven markets. - Incorrect

Incorrect Answer Feedback: Incorrect because dealers generally are unable or unwilling to hold these assets in their inventories, they will not make markets in them.

Question 5

Which of the following is most likely a consequence of the overconfidence bias? An investor who:

Correct answer: C

A. overestimates downside risks - Incorrect

Incorrect Answer Feedback: Incorrect because as a result of overconfidence bias, investors may underestimate, not overestimate, downside risks.

B. underestimates expected returns - Incorrect

Incorrect Answer Feedback: Incorrect because as a result of overconfidence bias investors may overestimate, not underestimate, expected returns.

C. holds a poorly diversified portfolio - Correct

Correct Answer Feedback: Correct because as a result of overconfidence bias, investors may underestimate risks and overestimate expected returns and/or hold poorly diversified portfolios, which may result in significant downside risk.

Question 6

Forward contracts are different from futures contracts because forwards are:

Correct answer: C

A. standardized. - Incorrect

Incorrect Answer Feedback: Incorrect because unlike futures contracts, forward contracts are not standardized.

B. marked to market. - Incorrect

Incorrect Answer Feedback: Incorrect because forward contracts are not marked to market like futures contracts.

C. traded over-the-counter. - Correct

Correct Answer Feedback: Correct because futures contracts differ from forward contracts in that they have standard terms, and are traded on a futures exchange, whereas forward contracts are traded over-the-counter.

Question 7

The market model is most likely used to predict:

Correct answer: C

A. market returns in a future period. - Incorrect

Incorrect Answer Feedback: Incorrect because the intercept, α, and slope coefficient, β, of i i the market model can be estimated by using historical security and market returns. These parameter estimates are then used to predict company-specific returns that a security may earn in a future period.

B. economic growth in a future period. - Incorrect

Incorrect Answer Feedback: Incorrect because the market model is a return generating model which can be used to predict company-specific returns based on market returns. Macroeconomic factor models (not the market model) use economic factors that are correlated with security returns. These factors may include economic growth.

C. company-specific returns in a future period. - Correct

Correct Answer Feedback: Correct because the intercept, α, and slope coefficient, β, of i i the market model can be estimated by using historical security and market returns. These parameter estimates are then used to predict company-specific returns that a security may earn in a future period.

Question 8

Which of the following ESG investment approaches best describes positive screening?

Correct answer: B

A. Systematic consideration of material ESG factors in asset allocation - Incorrect

Incorrect Answer Feedback: Incorrect because this describes ESG integration, not positive screening. ESG integration is defined as the systematic consideration of material ESG factors in asset allocation, security selection, and portfolio construction decisions.

B. Inclusion of sectors or companies in the portfolio based on specific ESG criteria - Correct

Correct Answer Feedback: Correct because positive screening is defined as including sectors or companies based on specific ESG criteria, typically ESG performance relative to industry peers.

C. Investments made with the intention to generate positive, measurable social and environmental impact - Incorrect

Incorrect Answer Feedback: Incorrect because it describes impact investing, not ESG integration. Impact investing is defined as investments made with the intention to generate positive, measurable social and environmental impact alongside a financial return.

Question 9

Which of the following is most likely undertaken during the risk budgeting process?

Correct answer: B

A. Assessing risk appetite - Incorrect

Incorrect Answer Feedback: Incorrect because it describes risk tolerance, not risk budgeting. Risk tolerance identifies the extent to which the organization is willing to experience losses or opportunity costs and to fail in meeting its objectives. Whereas risk tolerance focuses on the appetite for risk and what is and is not acceptable, risk budgeting has a more specific focus on how that risk is taken.

B. Setting a limit for value at risk (VaR) - Correct

Correct Answer Feedback: Correct because risk budgeting quantifies and allocates the tolerable risk by specific metrics. Four well-known single-dimension measures that are often used are standard deviation, beta, value at risk (VaR), and scenario loss.

C. Establishing a reserve to cover potential future losses - Incorrect

Incorrect Answer Feedback: Incorrect because it describes self-insurance, an example of risk acceptance, not risk budgeting. Self-insurance is the notion of bearing a risk that is considered undesirable but too costly to eliminate by external means. It may involve the establishment of a reserve to cover losses.

Question 10

An 8-year, 3.5% annual coupon bond is priced at 92.1492, with a yield to maturity of 4.7% and a Macaulay duration of 7.0705. If rates decrease by 75 bps, the percentage price change of the bond is closest to:

Correct answer: B

A. –5.30%. - Incorrect

Incorrect Answer Feedback: Incorrect because yields and bond prices are inversely related, the price change should be positive. Also, Macaulay duration, rather than modified duration, was used in the price change formula: 7.0705 x (–0.0075) = –0.05303 ≈ –5.3%.

B. 5.07%. - Correct

Correct Answer Feedback: Correct because to determine the percentage price change of a bond for a given change in yield, first convert Macaulay duration (7.0705) to modified duration by dividing Macaulay duration by 1 plus yield per period. ModDur = Next, multiply annual modified duration by the change in yield.

Correct Answer Feedback: Correct because to determine the percentage price change of a bond for a given change in yield, first convert Macaulay duration (7.0705) to modified duration by dividing Macaulay duration by 1 plus yield per period.

$$\text{ModDur} = \frac{7.0705}{1.047} = 6.7531$$

Next, multiply annual modified duration by the change in yield.

$$\%\Delta \text{PV}^{Full} \approx -6.7531 \times (-0.0075) = 0.0507$$

Figure 1: Source equation for question 10, option B

C. 5.30%. - Incorrect

Incorrect Answer Feedback: Incorrect because the price change is incorrectly calculated by using Macaulay duration, rather than modified duration. –7.0705 x (–0.0075) = – 0.05303 ≈ 5.3%.

Question 11

An analyst gathers the following information about a company (in $ thousands):

	Year 1	Year 2
Operating income	255	234
Depreciation and amortization	432	677
Interest expense	170	156

Based on the interest coverage ratio using EBITDA, the company's creditworthiness has:

Correct answer: C

A. deteriorated. - Incorrect

Incorrect Answer Feedback: Incorrect because over the period the interest coverage ratio increased and a higher ratio indicates higher, not lower, credit quality.

B. remained unchanged. - Incorrect

Incorrect Answer Feedback: Incorrect because EBIT (i.e., operating income) is used instead of EBITDA to calculate the interest coverage ratio. If the interest coverage ratio is calculated as EBIT/Interest expense, the ratios are: Year 1: 255/170 = 1.5x; Year 2: 234/156 = 1.5x. This stable ratio would imply no change to the company's creditworthiness.

C. improved. - Correct

Correct Answer Feedback: Correct because over the period the interest coverage ratio increased, which indicates that the company is better positioned to service its debt. A higher ratio indicates higher credit quality. Interest coverage ratio is calculated as EBITDA/Interest expense and EBITDA = Operating income + Depreciation and amortization, hence the ratios are: Year 1: EBITDA: 255 + 432 = 687 (cid:15482) Interest coverage: 687/170 = 4.04 ≈ 4.0x; Year 2: EBITDA: 234 + 677 = 911 (cid:15482) Interest coverage: 911/156 = 5.84 ≈ 5.8x.

Question 12

A correlation matrix of the returns for securities A, B, and C is reported below: Security A B C A 1 — — B 0.5 1 — C 0 -0.5 1 Assuming that the expected return and the standard deviation of each security are the same, a portfolio consisting of an equal allocation of which two securities will be most effective for portfolio diversification?

Correct answer: C

A. Securities A and B - Incorrect

Incorrect Answer Feedback: Incorrect. The neutral correlation of 0.0 between investment securities A and B means that these instruments have no correlation with each other. While portfolio A and B is superior to portfolio A and C, it is not superior to the negative correlation of portfolio B and C.

B. Securities A and C - Incorrect

Incorrect Answer Feedback: Incorrect. The positive correlation of 0.5 between investment securities A and C means that these instruments have positive correlation with each other and are not effective for portfolio diversification.

C. Securities B and C - Correct

Correct Answer Feedback: Correct. The negative correlation of –0.5 between investment securities B and C is the lowest and thus is the most effective for portfolio diversification.

Question 13

An analyst gathers the following information about a putable bond currently trading at par: Price per 100 of Par Value Change in Benchmark Yield 102.25 –25 bps 100.00 Unchanged 99.25 +25 bps The effective duration of this bond is:

Correct answer: C

A. 1.5. - Incorrect

Incorrect Answer Feedback: Incorrect because only the upward shift in the yield curve is used: EffDur ≠ ((PV)(PV)) / (2 * (ΔCurve) * PV) 0 + 0 = (100.00 – 99.25) / (2 * 0.0025 * 100.00) = 0.75 / (0.005 * 100) = 1.5

B. 4.5. - Incorrect

Incorrect Answer Feedback: Incorrect because only the increase in price is used: EffDur ≠ ((PV)(PV)) / (2 * (ΔCurve) * PV) – 0 0 = (102.25 – 100.00) / (2 * 0.0025 * 100.00) = 2.25 / (0.005 * 100) = 4.5

C. 6.0. - Correct

Correct Answer Feedback: Correct because effective duration is calculated as follows: EffDur = ((PV)(PV)) / (2 * (ΔCurve) * PV) – + 0 = (102.25 – 99.25) / (2 * 0.0025 * 100.00) = 3 / (0.005 * 100) = 6.0

Question 14

A private equity firm sells a portfolio company to a buyer that is active in the same industry as the portfolio company. This transaction is best described as a(n):

Correct answer: A

A. trade sale. - Correct

Correct Answer Feedback: Correct because in a trade sale, a portion or a division of the private company is sold either via direct sale or auction to a strategic buyer interested in increasing the scale and scope of the existing business.

B. secondary sale. - Incorrect

Incorrect Answer Feedback: Incorrect because this approach represents a sale of the company to another private equity firm or group of financial buyers.

C. initial public offering. - Incorrect

Incorrect Answer Feedback: Incorrect because an initial public offering involves the sale of shares to public investors. Public listing on an exchange can take place either as an initial public offering (IPO), a direct listing, or a special acquisition company (SPAC). IPOs are the most common means of raising capital in public equity markets using financial intermediaries to underwrite the offering. When a private equity firm or company founder takes a company public, the portfolio company sells its shares, including some or all of those held by the private equity firm, to public investors.

Question 15

Key rate duration:

Correct answer: B

A. requires the shape of the benchmark yield curve to stay constant. - Incorrect

Incorrect Answer Feedback: Incorrect because key rate duration assumes non-parallel shifts in the benchmark yield curve. Key rate duration provides insight into a bond’s sensitivity to non-parallel benchmark yield curve changes.

B. measures a bond's sensitivity to a change in the benchmark yield at a specific maturity. - Correct

Correct Answer Feedback: Correct because a key rate duration (or partial duration) is a measure of a bond's sensitivity to a change in the benchmark yield at a specific maturity.

C. indicates the same interest rate sensitivity as effective duration for non- parallel shifts in the yield curve. - Incorrect

Incorrect Answer Feedback: Incorrect because effective and key rate durations indicate the same sensitivity to interest rate changes when all yields change by the same amount. Effective duration measures a bond's sensitivity to changes in the benchmark yield curve if all yields change by the same amount. Key rate duration provides further insight into a bond's sensitivity to non-parallel benchmark yield curve changes. Key rate duration (or partial duration) is a measure of a bond's sensitivity to a change in the benchmark yield at a specific maturity. Key rate durations define a security's price sensitivity over a set of maturities along the yield curve, with the sum of key rate durations being equal to the effective duration when yields are shifted by the same amount.

Question 16

Which of the following natural resources has formed part of large institutional portfolios for decades?

Correct answer: B

A. Farmland only - Incorrect

Incorrect Answer Feedback: Incorrect because farmland as an investment is a more recent phenomenon, with only a few dedicated funds involved. With population growth, weather, and water management becoming more topical, however, investors may turn to these assets to actively address sustainability.

B. Timberland only - Correct

Correct Answer Feedback: Correct because timberland investment involves ownership of raw land and the harvesting of its trees for lumber, thus generating an income stream and the potential for capital gain, and timberland has been included in large institutional portfolios for decades. Farmland as an investment is a more recent phenomenon, with only a few dedicated funds involved. With population growth, weather, and water management becoming more topical, however, investors may turn to these assets to actively address sustainability.

C. Both farmland and timberland - Incorrect

Incorrect Answer Feedback: Incorrect because farmland as an investment is a more recent phenomenon, with only a few dedicated funds involved. With population growth, weather, and water management becoming more topical, however, investors may turn to these assets to actively address sustainability.

Question 17

According to put-call parity, a long position in an underlying is equivalent to a long risk-free bond combined with a:

Correct answer: B

A. long call and a long put with the same strike. - Incorrect

Incorrect Answer Feedback: Incorrect because by using the symbols and the signs in these versions of put–call parity, we can see several important interpretations, including long asset = long call, short put, long bond. Long call, long put and long bond will not be equivalent to the long asset.

B. long call and a short put with the same strike. - Correct

Correct Answer Feedback: Correct because by using the symbols and the signs in these versions of put–call parity, we can see several important interpretations, including long asset = long call, short put, long bond.

C. short call and a long put with the same strike. - Incorrect

Incorrect Answer Feedback: Incorrect because by using the symbols and the signs in these versions of put–call parity, we can see several important interpretations, including long asset = long call, short put, long bond. Short call, long put and long bond will not be equivalent to the long asset.

Question 18

An investor values a company based on the dividend discount model. All else being equal, the estimated intrinsic value of the stock for a one-year holding period is most likely:

Correct answer: B

A. lower than the estimated intrinsic value of the stock for a five-year holding period. - Incorrect

Incorrect Answer Feedback: Incorrect because the estimated intrinsic value of the stock for a one-year holding period will be the same as for a five-year holding period.

B. equal to the estimated intrinsic value of the stock for a five-year holding period. - Correct

Correct Answer Feedback: Correct because the general form of the dividend discount model (DDM) applies even in the case in which the investor has a finite investment horizon. For that investor, stock value today depends directly on the dividends the investor expects to receive before the stock is sold and depends indirectly on the expected dividends for periods subsequent to that sale, because those expected future dividends determine the expected selling price. Therefore, the estimated intrinsic value of the stock for a one-year holding period will be the same as for a five-year holding period.

C. higher than the estimated intrinsic value of the stock for a five-year holding period. - Incorrect

Incorrect Answer Feedback: Incorrect because the estimated intrinsic value of the stock for a one-year holding period will be the same as for a five-year holding period.

Question 19

Private real estate investors expect to generate the highest returns from:

Correct answer: C

A. core strategies. - Incorrect

Incorrect Answer Feedback: Incorrect because investors seeking higher returns than core strategies provide may also accept development, redevelopment, repositioning, and leasing risk. Finite-life closed-end funds are more commonly used for alpha- and beta-generating value-add and opportunistic investment styles. Value-add investments may require modest redevelopment or upgrades, the leasing of vacant space, or repositioning the underlying properties to earn a higher return than core properties. Opportunistic investments accept much higher risks than value-add or core strategies.

B. value-add strategies. - Incorrect

Incorrect Answer Feedback: Incorrect because investors seeking higher returns may also accept development, redevelopment, repositioning, and leasing risk. Finite-life closed-end funds are more commonly used for alpha- and beta-generating value-add and opportunistic investment styles. Value-add investments may require modest redevelopment or upgrades, the leasing of vacant space, or repositioning the underlying properties to earn a higher return than core properties.

C. opportunistic strategies. - Correct

Correct Answer Feedback: Correct because investors seeking higher returns may also accept development, re-development, repositioning, and leasing risk. Finite-life closed-end funds are more commonly used for alpha- and beta-generating value-add and opportunistic investment styles. Value-add investments may require modest redevelopment or upgrades, the leasing of vacant space, or repositioning the underlying properties to earn a higher return than core properties. Opportunistic investing accepts the much higher risks of development, major

redevelopment, repurposing of assets, taking on large vacancies, and speculating on significant improvement in market conditions.

Question 20

An investor purchased a 5-year bond at issuance for 98 per 100 of par value and held the bond for two years. Coupon payments were reinvested at the original yield to maturity. If the realized horizon yield is equal to the original yield to maturity, the investor most likely sold the bond at a price:

Correct answer: A

A. below par. - Correct

Correct Answer Feedback: Correct because the realized horizon yield matches the original yield-to-maturity if (1) coupon payments are reinvested at the same interest rate as the original yield-to-maturity, and (2) the bond is sold at a price on the constant-yield price trajectory, which implies that the investor does not have any capital gains or losses when the bond is sold. Since the bond is sold prior to maturity, the bond would have to be sold at a price greater than 98 but less than 100.

B. equal to par. - Incorrect

Incorrect Answer Feedback: Incorrect because if the bond was sold at par, the horizon yield would be greater than the yield-to-maturity.

C. above par. - Incorrect

Incorrect Answer Feedback: Incorrect because if the bond was sold above par, the horizon yield would be greater than the yield-to-maturity.

Question 21

For portfolios composed of the risk-free asset and a risky asset, the expected return as a function of standard deviation of returns is graphically represented by the:

Correct answer: B

A. security market line. - Incorrect

Incorrect Answer Feedback: Incorrect because the security market line (SML) is a graphical representation of the capital asset pricing model with beta, not standard deviation of returns, reflecting systematic risk, on the x-axis and expected return on the y-axis.

B. capital allocation line. - Correct

Correct Answer Feedback: Correct because a combination of the risk-free asset and a risky asset can result in a better risk–return trade-off than an investment in only one type of asset because the risk-free asset has zero correlation with the risky asset. The combination is called the capital allocation line.

C. Markowitz efficient frontier. - Incorrect

Incorrect Answer Feedback: Incorrect because the Markowitz efficient frontier is the curve that lies above and to the right of the global minimum-variance portfolio on the minimum- variance frontier, which in turn depicts portfolios with the minimum variance for a given return in a graph where the axes are standard deviation of returns and expected return. Thus, the Markowitz efficient frontier is a curve that contains many risky assets, and not the combinations of one risky asset with the risk-free asset.

Question 22

An investor gathers the following information about a company:

Most recent year's dividend per share	$1.47
Next year's estimate of earnings per share	$4.00
Estimate of long-run return on equity (ROE)	15%
Estimate of long-run dividend payout ratio	40%
Investors' required rate of return	12%

The company's justified forward P/E is closest to:

Correct answer: B

A. 10.0. - Incorrect

Incorrect Answer Feedback: Incorrect because there are two mistakes: (1) multiplying ROE with payout ratio instead of retention ratio when computing g, and (2) using retention ratio in the numerator instead of the payout ratio: g = Dividend payout ratio × ROE = 0.40 × 15 = 6% P /E = p/(r – g) = 0.60/(0.12 – 0.06) = 10 0 1

B. 13.3. - Correct

Correct Answer Feedback: Correct because Justified forward P/E: P /E = p/(r – g) 0 1 p = Payout ratio = 40% (given); r = Required rate of return = 12% g = (1 – Dividend payout ratio) × ROE = (1 – 0.40) × 15 = 9% P /E = p/(r – g) = 0.40/(0.12 – 0.09) = 13.3× 0 1 Alternatively: Justified forward P/E: P /E = (D /E)/(r – g) 0 1 1 1 g = (1 – Dividend payout ratio) × ROE = (1 – 0.40) × 15 = 9% D = $1.47 × 1.09 = 1.60; E = $4.00 (given); r = Required rate of return = 12% 1 1 (given) P /E = (D /E)/(r – g) = (1.60/4.00)/(0.12 – 0.09) = 13.3 0 1 1 1

C. 20.0. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses retention ratio instead of payout ratio in the numerator. P /E = 0.60/(0.12 – 0.09) = 20 0 1

Question 23

Which of the following categories of private debt most likely has the highest level of risk for an investor?

Correct answer: A

A. Mezzanine debt - Correct

Correct Answer Feedback: Correct because mezzanine debt is riskier than senior secure debt which is riskier than infrastructure debt (graphically displayed by the Private Capital Risk and Return Levels by Category chart). Because of its typically junior ranking and its usually unsecured status, mezzanine debt is riskier than senior secured debt. Investments in private capital vary in terms of risk and return across the corporate capital structure hierarchy. Typically, private equity, as the riskiest alternative, offers the highest returns, with private debt returns declining on a continuum down to the safest, most secured form of debt —infrastructure debt.

B. Senior direct debt - Incorrect

Incorrect Answer Feedback: Incorrect because senior direct lending is less risky than mezzanine debt. Because of its typically junior ranking and its usually unsecured status, mezzanine debt is riskier than senior secured debt.

C. Infrastructure debt - Incorrect

Incorrect Answer Feedback: Incorrect because investments in private capital vary in terms of risk and return across the corporate capital structure hierarchy. Typically, private equity, as the riskiest alternative, offers the highest returns, with private debt returns declining on a continuum down to the safest, most secured form of debt—infrastructure debt.

Question 24

A 5% decline in the value of a security held on margin reduces an investor's equity value by 20%. After the price decline, the leverage ratio of the position is:

Correct answer: B

A. 4.00. - Incorrect

Incorrect Answer Feedback: Incorrect because 1/25% = 4 and is the correct leverage ratio prior to the price decline. Once the price declines, leverage increases.

B. 4.75. - Correct

Correct Answer Feedback: Correct because the equity value decline is four times more than the security price, indicating initial leverage of 4. With a price decline of 5%, the total position value falls by 5%, however because of the 4 times leverage, equity falls by 20%. Accordingly, the new leverage ratio = 4 × (1 – 5%) / (1 – 20%) = 3.8 / 0.8 = 4.75.

C. 5.00. - Incorrect

Incorrect Answer Feedback: Incorrect because a leverage ratio of 5 results when the numerator portion of the calculation is not reduced by 5% (which is an incorrect calculation), i.e. new leverage ratio = 4 / (1 – 20%) = 4 / 0.8 = 5.

Question 25

A closed-end fund is trading at a premium to its NAV. This scenario most likely reflects:

Correct answer: C

A. concerns about the quality of management. - Incorrect

Incorrect Answer Feedback: Incorrect because concerns about the quality of management could suggest a discount from NAV.

B. excess demand for redemption of the shares. - Incorrect

Incorrect Answer Feedback: Incorrect because excess redemption demand suggests selling pressure, which could indicate a discount to NAV.

C. a belief that the portfolio securities are undervalued. - Correct

Correct Answer Feedback: Correct because closed-end funds may trade at a premium (discount) to NAV when investors believe that the portfolio securities are undervalued (overvalued).

Question 26

Which of the following US sovereign securities is a pure discount bond?

Correct answer: A

A. Treasury bill - Correct

Correct Answer Feedback: Correct because sovereign debt issues include the following: Short-term securities (with maturities ranging from 1 to 12 months), often known as Treasury bills, which are usually zero-coupon instruments sold at a discount to par.

B. Treasury note - Incorrect

Incorrect Answer Feedback: Incorrect because sovereign debt issues include the following: Short-term securities (with maturities ranging from 1 to 12 months), often known as Treasury bills, which are usually zero-coupon instruments sold at a discount to par. Medium- and long-term securities often known as Treasury notes and bonds. Fixed-rate coupon instruments in the domestic currency are most common, but sovereigns also issue floating- rate, inflation-linked, and foreign currency instruments.

C. Treasury bond - Incorrect

Incorrect Answer Feedback: Incorrect because sovereign debt issues include the following: Short-term securities (with maturities ranging from 1 to 12 months), often known as Treasury bills, which are usually zero-coupon instruments sold at a discount to par. Medium- and long-term securities often known as Treasury notes and bonds. Fixed-rate coupon instruments in the domestic currency are most common, but sovereigns also issue floating- rate, inflation-linked, and foreign currency instruments.

Question 27

An analyst gathers the following information about a company:

Earnings retention rate	55%
ROE	20%

If the required rate of return is 15%, the justified forward P/E of the company is closest to:

Correct answer: B

A. 7.5. - Incorrect

Incorrect Answer Feedback: Incorrect because in this case the dividend payout ratio is used during the calculation of the dividend growth rate instead of the earnings retention rate. The justified forward P/E can be calculated as where p is the dividend payout ratio, r is the required rate of return and g is the dividend growth rate. By definition earnings retention rate = (1 – Dividend payout ratio). It follows that p = 1 – earnings retention rate = 1 – 0.55 = 0.45. The dividend growth rate is incorrectly calculated as: g = earnings retention rate × ROE = 0.45 × 0.2 = 0.09. Thus, the justified forward P/E is p / (r – g) = 0.45 / (0.15 – 0.09) = 7.5.

Incorrect Answer Feedback: Incorrect because in this case the dividend payout ratio is used during the calculation of the dividend growth rate instead of the earnings retention rate. The justified forward P/E can be calculated as $\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{p}{r-g}$ where *p* is the dividend payout ratio, *r* is the required rate of return and *g* is the dividend growth rate. By definition earnings retention rate = (1 – Dividend payout ratio). It follows that *p* = 1 – earnings retention rate = 1 – 0.55 = 0.45. The dividend growth rate is incorrectly calculated as: *g* = earnings retention rate × ROE = 0.45 × 0.2 = 0.09. Thus, the justified forward P/E is *p* / (*r* – *g*) = 0.45 / (0.15 – 0.09) = 7.5.

Figure 2: Source equation for question 27, option A

B. 11.3. - Correct

Correct Answer Feedback: Correct because the justified forward P/E can be calculated as where p is the dividend payout ratio, r is the required rate of return and g is the dividend growth rate. By definition earnings retention rate = (1 – Dividend payout ratio). It follows that p = 1 – earnings retention rate = 1 – 0.55 = 0.45. The dividend growth rate is: g = earnings retention rate × ROE = 0.55 × 0.2 = 0.11. Thus, the justified forward P/E is p / (r – g) = 0.45 / (0.15 – 0.11) = 11.25.

Correct Answer Feedback: Correct because the justified forward P/E can be calculated as $\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{p}{r-g}$ where p is the dividend payout ratio, r is the required rate of return and g is the dividend growth rate. By definition earnings retention rate = (1 – Dividend payout ratio). It follows that p = 1 – earnings retention rate = 1 – 0.55 = 0.45. The dividend growth rate is: g = earnings retention rate × ROE = 0.55 × 0.2 = 0.11. Thus, the justified forward P/E is p / (r – g) = 0.45 / (0.15 – 0.11) = 11.25.

Figure 3: Source equation for question 27, option B

C. 13.8. - Incorrect

Incorrect Answer Feedback: Incorrect because in this case the earnings retention rate is used instead of the dividend payout ratio in the formula for the justified forward P/E, which is: where p is the dividend payout ratio, r is the required rate of return and g is the dividend growth rate. In this case we use p = 0.55 instead of p = 1 – 0.55 = 0.45. The dividend growth rate is: g = earnings retention rate × ROE = 0.55 × 0.2 = 0.11. Thus, the justified forward P/E is p / (r – g) = 0.55 / (0.15 – 0.11) = 13.75.

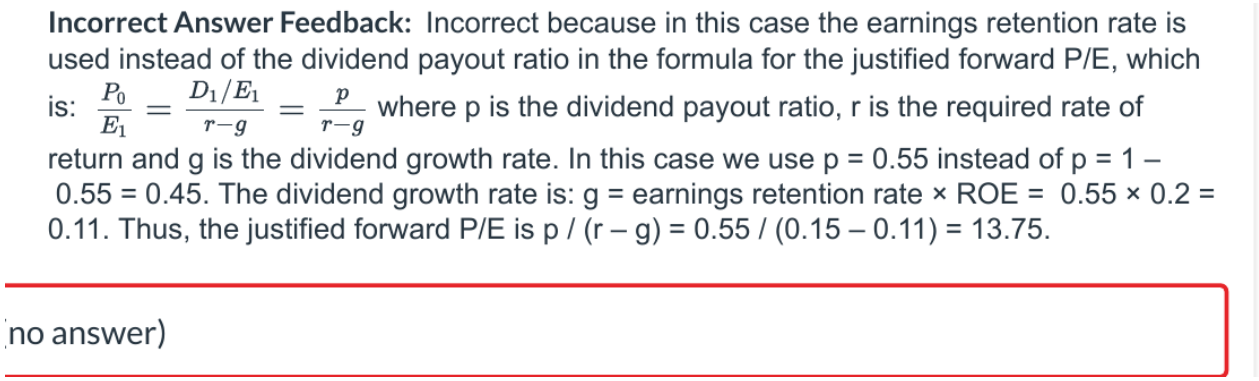


Figure 4: Source equation for question 27, option C

Question 28

The pass-through rate of a mortgage-pass through security is:

Correct answer: A

A. lower than the mortgage rate on the underlying pool. - Correct

Correct Answer Feedback: Correct because the pass-through rate is lower than the mortgage rate on the underlying pool of mortgages by an amount equal to the servicing and other administrative fees.

B. equal to the mortgage rate on the underlying pool. - Incorrect

Incorrect Answer Feedback: Incorrect because the pass-through rate is lower than the mortgage rate on the underlying pool of mortgages by an amount equal to the servicing and other administrative fees.

C. greater than the mortgage rate on the underlying pool. - Incorrect

Incorrect Answer Feedback: Incorrect because the pass-through rate is lower than the mortgage rate on the underlying pool of mortgages by an amount equal to the servicing and other administrative fees.

Question 29

Consider a call option trading for $2.00 with an exercise price of $38.00. If the price of the underlying at expiration is $42.00, the payoff for the call option seller is:

Correct answer: A

A. –$4.00. - Correct

Correct Answer Feedback: Correct because the call seller's payoff as: –c = –Max(0, S – T T X) = –Max(0, $42.00 – $38.00) = –$4.00.

B. –$2.00. - Incorrect

Incorrect Answer Feedback: Incorrect because the call seller's profit (not payoff) is calculated as: –Max(0, S – X) + c = –Max(0, $42.00 – $38.00) + $2.00 = –$2.00. T 0

C. $4.00. - Incorrect

Incorrect Answer Feedback: Incorrect because the call buyer's (not seller's) payoff is calculated as: Max(0, S – X) = Max(0, $42.00 – $38.00) = $4.00. T

Question 30

A credit issuer rating assigned by a rating agency is based on:

Correct answer: C

A. the notching adjustment of a specific bond of the issuer. - Incorrect

Incorrect Answer Feedback: Incorrect because an issue rating is notched from the issuer rating, not the other way round. The rating agencies have adopted a notching process whereby their credit ratings on issues can be moved up or down from the issuer rating,

B. the ranking of a specific bond in the issuer's capital structure. - Incorrect

Incorrect Answer Feedback: Incorrect because an issue rating, not an issuer rating, should consider ranking in the capital structure. Issue ratings refer to specific financial obligations of an issuer and take into consideration such factors as ranking in the capital structure (e.g., secured or subordinated).

C. the issuer's ability and willingness to service its senior unsecured debt. - Correct

Correct Answer Feedback: Correct because an issuer credit rating is meant to address an obligor's overall creditworthiness — its ability and willingness to make timely payments of interest and principal on its debt. The issuer credit rating usually applies to its senior unsecured debt.

Question 31

The price sensitivity of a bond to a change in the benchmark yield curve is the:

Correct answer: A

A. effective duration. - Correct

Correct Answer Feedback: Correct because an approach to assess the interest rate risk of a bond is to estimate the percentage change in price given a change in benchmark yield curve - for example, the government par curve. This estimate, which is very similar to the formula for approximate modified duration, is called the effective duration. The effective duration of a bond is the sensitivity of the bond's price to a change in a benchmark yield curve. Effective duration is essential to the measurement of the interest rate risk of a complex bond, such as a bond that contains an embedded call option. Therefore, effective duration is the best duration measure of the change to a bond's price given a change in a benchmark yield curve.

B. modified duration. - Incorrect

Incorrect Answer Feedback: Incorrect because the calculation of the modified duration (ModDur) statistic of a bond requires a simple adjustment to Macaulay duration. It is the Macaulay duration statistic divided by one plus the yield per period. Although modified duration might seem to be just a Macaulay duration with minor adjustments, it has an important application in risk management: Modified duration provides an estimate of the percentage price change for a bond given a change in its yield-to-maturity. Effective duration is the best duration measure for a given change in benchmark yield and not modified duration which is the best measure for a given change in a bond's yield-to-maturity.

C. Macaulay duration. - Incorrect

Incorrect Answer Feedback: Incorrect because Macaulay duration is a weighted average of the time to receipt of the bonds promised payments, where the weights are the shares of the full price that correspond to each of the bond's promised payments.

Question 32

A transaction setup to absorb the variable cash flow of a floating-rate asset can be described as a:

Correct answer: B

A. fair value hedge. - Incorrect

Incorrect Answer Feedback: Incorrect because a fair value hedge designation applies when a derivative is deemed to offset the fluctuation in fair value of an asset or liability. A fair value hedge type is not aimed at absorbing variable cash flow of a floating-rate asset.

B. cash flow hedge. - Correct

Correct Answer Feedback: Correct because derivatives designated as absorbing the variable cash flow of a floating-rate asset or liability such as foreign exchange, interest rates, or commodities are referred to as cash flow hedges.

C. net investment hedge. - Incorrect

Incorrect Answer Feedback: Incorrect because net investment hedges occur when either a foreign currency bond or a derivative such as an FX swap or forward is used to offset the exchange rate risk of the equity of a foreign operation. A net investment hedge type is not aimed at absorbing variable cash flow of a floating-rate asset.

Question 33

Measuring the duration of a bond portfolio as a weighted average of the individual bond durations:

Correct answer: B

A. is difficult to use as a measure of interest rate risk. - Incorrect

Incorrect Answer Feedback: Incorrect because the weighted average of the individual bond durations that comprise the portfolio is easily used as a measure of interest rate risk.

B. assumes the yield curve shifts in a parallel manner. - Correct

Correct Answer Feedback: Correct because the weighted average of the individual bond durations that comprise the portfolio implicitly assumes a parallel shift in the yield curve. A parallel yield curve shift implies that all rates change by the same amount in the same direction. In reality, interest rate changes frequently result in a steeper or flatter yield curve.

C. assumes the yield curve shifts in a non-parallel manner. - Incorrect

Incorrect Answer Feedback: Incorrect because the weighted average of the individual bond durations that comprise the portfolio implicitly assumes a parallel shift in the yield curve. A parallel yield curve shift implies that all rates change by the same amount in the same direction. In reality, interest rate changes frequently result in a steeper or flatter yield curve.

Question 34

An analyst gathers the following information (in $ millions) about a private equity fund:

	Investment A	Investment B	Total
Initial cost	100	150	250
Profit/loss	–50	30	–20

The manager's incentive fee is 20% of the profits. If no clawback provision applies, the total incentive fee (in $ millions) under an American-style waterfall is:

Correct answer: C

A. –4. - Incorrect

Incorrect Answer Feedback: Incorrect because it calculated the incentive fee based on the total loss of –$20 million. Note that under a European-style whole-of-fund waterfall, while the incentive fee is calculated on the total profit/loss of the fund, the minimum incentive fee would be $0 instead of –$20 million × 20% = –$4 million.

B. 0. - Incorrect

Incorrect Answer Feedback: Incorrect because it calculated the incentive fee based on the total loss of –$20 million under a European-style whole-of-fund waterfall instead of an American-style deal-by-deal waterfall. Consequently, total incentive fee payable = MAX [$0, –$4 million] = $0. Note that while the incentive fee is calculated on the total profit/loss of the fund, the minimum incentive fee is $0 instead of –$20 million × 20% = –$4 million.

C. 6. - Correct

Correct Answer Feedback: Correct because it calculated the incentive fee based on the Investment B's profit of $30 million under an American-style deal-by-deal waterfall, and correctly excluded Investment A which incurred a loss. Total incentive fee payable = Incentive fee of Investment B = $30 million × 20% = $6 million.

Question 35

Selected information about shares of two companies is provided in the following table: Stock Standard Deviation Correlation of Returns* Portfolio Weights Cable Incorporated 30% 0.65 68% GPT Company 20% 0.65 32% * Correlation of returns between Cable Incorporated and GPT Company. The standard deviation of returns of the portfolio formed with these two stocks is closest to:

Correct answer: A

A. 25.04%. - Correct

Correct Answer Feedback: Correct because the portfolio standard deviation for a two stock portfolio, σ_p , is calculated as follows: $\sigma_p = [w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1 w_2 \rho_{12}\sigma_1\sigma_2]^{1/2}$ Where w = weight of each stock σ = standard deviation ρ = the correlation of returns between Stock 1 and Stock 2 12 Plugging in the numbers $\sigma_p = [(0.68)^2(0.3)^2+(0.32)^2(0.2)^2+2(0.68)(0.32)(0.65)(0.3(0.2)] = 0.2504$.

B. 26.80%. - Incorrect

Incorrect Answer Feedback: Incorrect. This incorrect calculation is the weighted standard deviations = 0.68 × 0.30 + 0.32 × 0.20 = 0.268.

C. 32.85%. - Incorrect

Incorrect Answer Feedback: Incorrect. The incorrect calculation excludes the standard deviation from the third part of the equation; it also does not take 2 2 2 2 the square root. This is incorrect calculation is [(0.68)(0.3) + (0.32)(0.2) + 2(0.68)(0.32)(0.65) = 0.3285.

Question 36

Market prices are most likely accepted as accurately reflecting intrinsic values in:

Correct answer: B

A. an inefficient market only. - Incorrect

Incorrect Answer Feedback: Incorrect because if investors believe an asset market is relatively inefficient, they may try to develop an independent estimate of intrinsic value, indicating market prices are not likely to be accepted as reflecting intrinsic values.

B. a highly efficient market only. - Correct

Correct Answer Feedback: Correct because if investors believe a market is highly efficient, they will usually accept market prices as accurately reflecting intrinsic values.

C. both an inefficient market and a highly efficient market. - Incorrect

Incorrect Answer Feedback: Incorrect because if investors believe an asset market is relatively inefficient, they may try to develop an independent estimate of intrinsic value, indicating market prices are not likely to be accepted as reflecting intrinsic values.

Question 37

Sponsored depository receipts are most likely subject to:

Correct answer: C

A. less reporting requirements than unsponsored ones. - Incorrect

Incorrect Answer Feedback: Incorrect because sponsored DRs are generally subject to greater reporting requirements than unsponsored DRs.

B. the same level of reporting requirements as unsponsored ones. - Incorrect

Incorrect Answer Feedback: Incorrect because sponsored DRs are generally subject to greater reporting requirements than unsponsored DRs.

C. greater reporting requirements than unsponsored ones. - Correct

Correct Answer Feedback: Correct because sponsored DRs are generally subject to greater reporting requirements than unsponsored DRs.

Question 38

All else being equal, which of the following conditions most likely results in wider credit spreads?

Correct answer: C

A. A credit cycle peak - Incorrect

Incorrect Answer Feedback: Incorrect because spreads are tightest at or near the top of the credit cycle.

B. A strengthening economy - Incorrect

Incorrect Answer Feedback: Incorrect because a strengthening economy will cause credit spreads to narrow.

C. A period of heavy new issue supply - Correct

Correct Answer Feedback: Correct because in periods of heavy new issue supply, credit spreads will widen if there is insufficient demand. In periods of high demand for bonds, spreads will move tighter.

Question 39

Which of the following is most likely an application of the CAPM?

Correct answer: B

A. Estimating expected returns over multiple periods - Incorrect

Incorrect Answer Feedback: Incorrect because a limitation of the CAPM is that it is a single-period model that does not consider multi-period implications or investment objectives of future periods, which can lead to myopic and suboptimal investment decisions.

B. Assessing return performance against a benchmark - Correct

Correct Answer Feedback: Correct because applications of the CAPM include comparison of the actual return of a portfolio or portfolio manager with the CAPM return for performance appraisal.

C. Estimating expected returns using multiple investment factors - Incorrect

Incorrect Answer Feedback: Incorrect because a limitation of the CAPM is that it is a single-factor model. Only systematic risk or beta risk is priced in the CAPM. Thus, the CAPM states that no other investment characteristics should be considered in estimating returns.

Question 40

Which of the following statements about market efficiency is most accurate? In a:

Correct answer: C

A. weak-form efficient market, technical analysis generates abnormal returns over the long run. - Incorrect

Incorrect Answer Feedback: Incorrect because securities markets are weak-form efficient, and therefore, investors cannot earn abnormal returns by trading on the basis of past trends in price (using technical analysis).

B. semi-strong-form efficient market, active portfolio managers outperform the market on a consistent basis. - Incorrect

Incorrect Answer Feedback: Incorrect because if securities markets are weak-form and semi-strong-form efficient, the implication is that active trading, whether attempting to exploit price patterns or public information, is not likely to generate abnormal returns. In other words, active portfolio managers cannot beat the market on a consistent basis.

C. strong-form efficient market, insiders would not be able to earn abnormal returns from trading on private information. - Correct

Correct Answer Feedback: Correct because in the case of a strong-form efficient market, insiders would not be able to earn abnormal returns from trading on the basis of private information.

Question 41

If the 1-year spot rate is 3% and the 2-year spot rate is 4.5%, the 1y1y implied forward rate is closest to:

Correct answer: C

A. 1.46%. - Incorrect

Incorrect Answer Feedback: Incorrect because the exponents are missing in the formula. ≠ (1 + z) × (1 + IFR,) = (1 + z) A A,B–A B = (1 + 0.03) × (1 + IFR,) = (1 + 4.5) A,B–A =1.045/1.03 = 1.01456 ≈ 1.46%.

B. 3.75%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is simply the average of the one and two year spot rates, not the implied forward rate = (3% + 4.5%) / 2 = 3.75%.

C. 6.02%. - Correct

Correct Answer Feedback: Correct because the implied forward rate can be calculated as: (1 + z) A × (1 + IFR,) B–A = (1 + z) B, where: A A,B–A B z = the 1 year spot rate A z = the 2 year spot rate IFR = the implied forward rate IFR = (1 + 0.03) × (1 + IFR,) B– B A,B–A A = (1 + 0.045)2 2 = 1.045 /1.03 = 1.0602 ≈ 6.02%.

Question 42

Exercise of a European put option is most likely justified if:

Correct answer: C

A. the option is out of the money. - Incorrect

Incorrect Answer Feedback: Incorrect because an out-of-the-money option should not be exercised and will expire worthless.

B. the exercise value is negative. - Incorrect

Incorrect Answer Feedback: Incorrect because an option that generates a negative cash flow when exercised should not be exercised.

C. the exercise price exceeds the value of the underlying. - Correct

Correct Answer Feedback: Correct because if the exercise price exceeds the value of the underlying at expiration, the option has positive exercise value and may be exercised.

Question 43

If a stock purchased with 60% margin declines 25%, the investor's return on equity is closest to:

Correct answer: C

A. –85%. - Incorrect

Incorrect Answer Feedback: Incorrect because 85% is the sum of 60% and 25%, the two numbers provided in the stem, and is not a meaningful number or calculation.

B. –63%. - Incorrect

Incorrect Answer Feedback: Incorrect because if 40% rather than 60% is incorrectly used to calculate leverage, leverage = 1/0.4 ≈ 2.5, multiplied by the 25% decline ≈ a 63% decline.

C. –42%. - Correct

Correct Answer Feedback: Correct because 60% margin equals leverage of 1/0.6 ≈ 1.67, multiplied by the 25% decline ≈ a 42% decline.

Question 44

Do management fees get paid to the manager of a hedge fund regardless of the fund's performance?

Correct answer: A

A. Yes - Correct

Correct Answer Feedback: Correct because regardless of performance, the management fee is always paid to the fund manager.

B. No, only when the fund's gross return is positive - Incorrect

Incorrect Answer Feedback: Incorrect because the gross return can be at any level and the manager is still paid the management fee.

C. No, only when the fund's net asset value exceeds the previous high-water mark - Incorrect

Incorrect Answer Feedback: Incorrect because the management fee is paid regardless of the value of the assets in the fund.

Question 45

Which of the following equations best describes put–call parity for European options? T

Correct answer: A

A. p = c − S + X/(1 + r) 0 0 0 - Correct

Correct Answer Feedback: Correct because S + p = c + X / (1 + r)T. This relationship 0 0 0 is known as put-call parity. Rearranging the equation, we can get: p = c − S + X / (1 + 0 0 0 r)T. That is, long put = long call, short asset, long bond. T

B. c = p − S − X/(1 + r) 0 0 0 - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the response rationale for the correct answer, according to put-parity, S + p = c + X / (1 + r)T. Rearranging the put- 0 0 0 call parity equation, we get c = p +S − X / (1 + r)T. Therefore, c ≠ p −S − X / (1 + r)T. 0 0 0 0 0 0 T

C. S = c − p − X/(1 + r) 0 0 0 - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the response rationale for the correct answer, according to put-call parity, S + p = c + X / (1 + r)T. Rearranging, we 0 0 0 can get S = c − p + X / (1 + r)T. Therefore, S ≠ c − p − X / (1 + r)T. 0 0 0 0 0 0

Question 46

In the binomial option pricing model, the price of a call option will increase when:

Correct answer: B

A. the risk-free rate increases. - Incorrect

Incorrect Answer Feedback: Incorrect because the value of call option = c = 0 + – (πc + (1 – π)c)/(1 + r), where π = (1 + r – d)/(u – d), the formula takes the 1 1 form of an expected future value, the numerator, discounted at the risk-free rate. Since the expected future value is discounted at the risk-free rate, a higher risk-free rate will result in a lower call option price.

B. the volatility of the underlying increases. - Correct

Correct Answer Feedback: Correct because if volatility increases, the difference between S + and S − increases, which widens the range between c + and c −, leading to a higher 1 1 1 1 option value.

C. the probability of an upward movement in the underlying increases. - Incorrect

Incorrect Answer Feedback: Incorrect because the value of call option = c = 0 + – (πc + (1 – π)c)/(1 + r), where π = (1 + r – d)/(u – d), the probabilities of the 1 1 up and down moves, q and 1 – q, do not appear in the formula. The call option price is effectively independent of the up move probability.

Question 47

Issuers' disclosures about operating costs are most likely:

Correct answer: A

A. less detailed than their revenue disclosures. - Correct

Correct Answer Feedback: Correct because issuers' disclosures about operating costs are typically less detailed than revenue disclosures. Rather than modeling costs separately for different geographic regions, business segments, or product lines, analysts are often forced to use more aggregated forecast objects such as consolidated financial statement lines (e.g., cost of sales, SG&A) or summary measures like EBITDA margins on a consolidated or segment basis.

B. just as detailed as their revenue disclosures. - Incorrect

Incorrect Answer Feedback: Incorrect because issuers' disclosures about operating costs are typically less detailed than, not as detailed as, revenue disclosures.

C. more detailed than their revenue disclosures. - Incorrect

Incorrect Answer Feedback: Incorrect because issuers' disclosures about operating costs are typically less detailed than revenue disclosures.

Question 48

The industry growth style of the crude oil sector is best characterized as:

Correct answer: A

A. cyclical mature. - Correct

Correct Answer Feedback: Correct because crude oil together with natural gas and freight transportation are examples of cyclical mature industries given in the reading.

B. defensive growth. - Incorrect

Incorrect Answer Feedback: Incorrect because biotechnology, software and gaming are examples of defensive growth industries given in the reading, not crude oil which is a cyclical mature industry instead.

C. defensive mature. - Incorrect

Incorrect Answer Feedback: Incorrect because utilities, beverages and pharmaceuticals are examples of defensive mature industries given in the reading, not crude oil which is a cyclical mature industry instead.

Question 49

In which section of an IPS is the rebalancing policy most likely found?

Correct answer: A

A. Appendices - Correct

Correct Answer Feedback: Correct because Appendices details (A) Strategic Asset Allocation (B) Rebalancing Policy. Many investors specify a strategic asset allocation (SAA), also known as the policy portfolio, which is the baseline allocation of portfolio assets to asset classes in view of the investor's investment objectives.

B. Investment Guidelines - Incorrect

Incorrect Answer Feedback: Incorrect because Investment Guidelines provides information about how policy should be executed (e.g., on the permissible use of leverage and derivatives) and on specific types of assets excluded from investment, if any. It is Appendices that details (A) Strategic Asset Allocation (B) Rebalancing Policy.

C. Evaluation and Review - Incorrect

Incorrect Answer Feedback: Incorrect because Evaluation and Review provides guidance on obtaining feedback on investment results. It is Appendices that details (A) Strategic Asset Allocation (B) Rebalancing Policy.

Question 50

For a given currency and tenor, a bond's yield spread over the standard swap rate is best described as the:

Correct answer: A

A. I-spread. - Correct

Correct Answer Feedback: Correct because the yield spread of a specific bond over the standard swap rate in that currency of the same tenor is known as the I-spread or interpolated spread to the swap curve. This yield spread over MRR allows comparison of bonds with differing credit and liquidity risks against an interbank lending benchmark.

B. G-spread. - Incorrect

Incorrect Answer Feedback: Incorrect because the yield spread in basis points over an actual or interpolated government bond is known as the G-spread. The spread over a government bond is the return for bearing greater credit, liquidity, and other risks relative to the sovereign bond.

C. Z-spread. - Incorrect

Incorrect Answer Feedback: Incorrect because the G-spread and I-spread each use the same discount rate for each cash flow. Another approach is to calculate a constant yield spread over a government (or interest rate swap) spot curve instead. This spread is known as the zero-volatility spread (Z-spread) of a bond over the benchmark rate.

Question 51

The minimum rate of return required by investors to purchase an equity security is most likely approximated by the company's:

Correct answer: A

A. cost of equity. - Correct

Correct Answer Feedback: Correct because the company's cost of equity is often used as a proxy for the investors' minimum required rate of return.

B. return on equity. - Incorrect

Incorrect Answer Feedback: Incorrect because when investors purchase the company's equity securities, their minimum required rate of return is based on the future cash flows they expect to receive. Return on equity is a historical accounting measure which measures the total return on the firm's book value, therefore it may or may not approximate investors' minimum required rate of return.

C. weighted average cost of capital. - Incorrect

Incorrect Answer Feedback: Incorrect because the WACC represents the minimum required rate of return that the company must earn on its long-term investments to satisfy all providers of capital, i.e. both debt holders and equity investors not just equity investors.

Question 52

When defining an asset class for the purpose of asset allocation, an investor should ensure that assets in the class are relatively:

Correct answer: A

A. homogeneous and have low paired correlations with other asset classes. - Correct

Correct Answer Feedback: Correct because an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes. In statistical terms, risk and return expectations should be similar and paired correlations of asset should be relatively high within an asset class but should be lower versus assets in other asset classes.

B. homogeneous and have high paired correlations with other asset classes. - Incorrect

Incorrect Answer Feedback: Incorrect because an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes. In statistical terms, risk and return expectations should be similar and paired correlations of asset should be relatively high within an asset class but should be lower versus assets in other asset classes.

C. heterogeneous and have low paired correlations with other asset classes. - Incorrect

Incorrect Answer Feedback: Incorrect because an asset class should contain relatively homogeneous assets while providing diversification relative to other asset classes. In statistical terms, risk and return expectations should be similar and paired correlations of asset should be relatively high within an asset class but should be lower versus assets in other asset classes.

Question 53

For an option-free, fixed-rate bond the coupon reinvestment risk most likely dominates market price risk when the investment horizon is:

Correct answer: C

A. less than the Macaulay duration of the bond. - Incorrect

Incorrect Answer Feedback: Incorrect because for an option-free, fixed-rate bond when the investment horizon is greater than the Macaulay duration, the coupon reinvestment risk will tend to dominate market price risk.

B. equal to the Macaulay duration of the bond. - Incorrect

Incorrect Answer Feedback: Incorrect because for an option-free, fixed-rate bond when the investment horizon is greater than the Macaulay duration, the coupon reinvestment risk will tend to dominate market price risk.

C. greater than the Macaulay duration of the bond. - Correct

Correct Answer Feedback: Correct because for an option-free, fixed-rate bond when the investment horizon is greater than the Macaulay duration, the coupon reinvestment risk will tend to dominate market price risk. In such situations, the investor's risk is to lower interest rates.

Question 54

An investor owns a stock trading at $22 per share and would like to place an order to sell the position should the stock ever trade at or below $20 per share. The order most likely is:

Correct answer: B

A. limit 20 sell. - Incorrect

Incorrect Answer Feedback: Incorrect because this order will sell the stock if it trades at or above $20/share, but not if it trades below $20/share.

B. stop 20, market sell. - Correct

Correct Answer Feedback: Correct because this order will sell the position at $20/share or under, at all prices, at the market. If the price falls to or below $20, the market order becomes valid and it should immediately execute at the best price then available in the market.

C. stop 20, limit 20 sell. - Incorrect

Incorrect Answer Feedback: Incorrect because this order will sell the stock at $20/share, but not if it trades below/through $20/share.

Question 55

Which of the following most likely has the greatest impact on portfolio performance?

Correct answer: A

A. The weighting of asset classes - Correct

Correct Answer Feedback: Correct because decisions that need to be made in the asset allocation of the portfolio include the distribution between equities, fixed-income securities, and cash; sub-asset classes, such as corporate and government bonds; and geographical weightings within asset classes. Although all decisions have an effect on portfolio performance, the asset allocation decision is commonly viewed as having the greatest impact.

B. The selection of securities within sectors - Incorrect

Incorrect Answer Feedback: Incorrect because although all decisions have an effect on portfolio performance, the asset allocation decision is commonly viewed as having the greatest impact. The selection of securities is not an asset allocation decision.

C. The trading of securities within asset classes - Incorrect

Incorrect Answer Feedback: Incorrect because although all decisions have an effect on portfolio performance, the asset allocation decision is commonly viewed as having the greatest impact. The trading of securities is not an asset allocation decision.

Question 56

All else being equal, the effect of a cash dividend on shareholders' wealth is most likely equivalent to the effect of a:

Correct answer: C

A. stock split on shareholders' wealth. - Incorrect

Incorrect Answer Feedback: Incorrect because stock splits and reverse stock splits are similar to stock (not cash) dividends in that they have no economic effect on the company or shareholders.

B. stock dividend on shareholders' wealth. - Incorrect

Incorrect Answer Feedback: Incorrect because stock splits and reverse stock splits are similar to stock (not cash) dividends in that they have no economic effect on the company or shareholders.

C. stock repurchase on shareholders' wealth. - Correct

Correct Answer Feedback: Correct because a share repurchase is viewed as equivalent to the payment of cash dividends of equal value in terms of the effect on shareholders' wealth, all other things being equal.

Question 57

Which of the following statements regarding a commodity index is most accurate?

Correct answer: C

A. Commodity indexes commonly use an equal-weighting method. - Incorrect

Incorrect Answer Feedback: Incorrect because commodity indexes do not have an obvious weighting mechanism; therefore, commodity index providers create their own weighting methods.

B. Commodity indexes in the same markets will share similar risk and return profiles. - Incorrect

Incorrect Answer Feedback: Incorrect because different weighting methods lead to different exposure to specific commodities and result in different risk and return profiles. Unlike commodity indexes, broad equity and fixed-income indexes that target the same markets share similar risk and return profiles.

C. Commodity index returns differ from the changes in the prices of their underlying commodities. - Correct

Correct Answer Feedback: Correct because the performance of commodity indexes can be different from their underlying commodities because the indexes consist of futures contracts on the commodities rather than the actual commodities. Commodity index returns reflect the risk-free interest rate, the changes in future prices, and the roll yield.

Question 58

An analyst gathers the following information about a portfolio and the market:

	Return	Standard Deviation of Returns
Portfolio	11%	18%
Market	10%	20%

2 If the risk-free rate is 4%, the portfolio's M alpha is closest to:

Correct answer: C

A. 0.3%. - Incorrect

Incorrect Answer Feedback: Incorrect because the standard deviation ratio is inverted: M2 alpha = (R – R) × (σ /σ) + R – R p f p f m m 2 M alpha = (0.11 – 0.04) × (0.18/0.20) + 0.04 – 0.10 = 0.07 × 0.90 – 0.06 = 0.3%. Alternatively, it applies the ratio of standard deviations to the market risk premium rather than the portfolio's risk premium: M2 alpha = (R – R) – (R – R) × (σ /σ) p f m f m p 2 M alpha = (0.11 – 0.04) – (0.10 – 0.04) × (0.20/0.18) = 0.07 – 0.06 × 1.1111 = 0.333% ≈ 0.3%.

B. 0.8%. - Incorrect

Incorrect Answer Feedback: Incorrect because the market risk premium is replaced with the portfolio risk premium in the last section of the formula. M2 alpha = (R – R) × (σ /σ) – (R – R) p f m p f p 2 M alpha = (0.11 – 0.04) × (0.20/0.18) – (0.11 – 0.04) = 0.07 × 1.1111 – 0.07 = 0.778% ≈ 0.8%.

C. 1.8%. - Correct

Correct Answer Feedback: Correct because the difference between the risk-adjusted performance (M2) of the portfolio and the performance of the market is frequently referred to as M2 alpha. Applying the formula yields: 2 M alpha = (R – R) × (σ /σ) + R – R p f m f m p M2 alpha = (0.11 – 0.04) × (0.20/0.18) + 0.04 – 0.10 = 0.07 × 1.1111 – 0.06 = 1.778% ≈ 1.8%.

Question 59

When developing an IPS, an investor's desire to earn an inflation-adjusted annual rate of return of at least 2% is best classified as a(n):

Correct answer: C

A. relative risk objective. - Incorrect

Incorrect Answer Feedback: Incorrect because the objective provided in the stem is a return objective, and not a risk objective.

B. relative return objective. - Incorrect

Incorrect Answer Feedback: Incorrect because the return objective is formulated in absolute terms, even though it is stated in real (inflation-adjusted) terms. Alternatively, the return objective can be stated on a relative basis—for example, relative to a benchmark return, which is not the case in the stem.

C. absolute return objective. - Correct

Correct Answer Feedback: Correct because return objectives may be stated on an absolute or a relative basis. As an example of an absolute objective, the client may want to achieve a particular percentage rate of return. This objective could be a nominal rate of return or could be expressed in real (inflation-adjusted) terms.

Question 60

If a forward contract requires no cash outlay at initiation, it is most likely true that at initiation:

Correct answer: B

A. value exceeds price. - Incorrect

Incorrect Answer Feedback: Incorrect. Value is zero; price is a positive number.

B. price exceeds value. - Correct

Correct Answer Feedback: Correct. At initiation, value is equal to zero. Price is a positive number that states the amount that must be paid when the purchase takes place.

C. price is equal to value. - Incorrect

Incorrect Answer Feedback: Incorrect. Value is zero; price is a positive number.

Question 61

An analyst gathers the following information about a company's common stock and the market: Dividend per share (D) $3.00 0 Dividend payout ratio 25%

Equity beta	1.6
Expected risk-free rate of return	3.5%
Expected market rate of return	9.0%
ROE	7.0%

Using the Gordon growth model, the stock's intrinsic value is closest to:

Correct answer: C

A. $24.96. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer incorrectly calculates the required rate of return. Instead of applying the CAPM formula as: r = R + β[E(R) – R], the F i M F formula is incorrectly applied as: r = R + β[E(R)]. That is, r = 3.5% + 1.6 × (9.0%) = 17.90%. F i M Therefore, applying this incorrect r into the formula: V = D (1 + g)/(r – g) = D /(r – g), the 0 0 1 candidate mistakenly arrives at: V = 3.1575/(17.90% – 5.25%) = $24.96, where g = b × ROE, 0 where b = earnings retention rate = (1 – dividend payout ratio). Therefore, b = (1 – 0.25) = 0.75. Therefore, g = 0.75 × 7% = 5.25%.

B. $42.55. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer incorrectly applies D0 (instead of D) in the formula V = D (1 + g)/(r – g) = D /(r – g). That is: 1 0 0 1 Step 1: calculate g, where: g = b × ROE, where b = earnings retention rate = (1 – dividend payout ratio). Therefore, b = (1 – 0.25) = 0.75. Therefore, g = 0.75 × 7% = 5.25%. Step 2: calculate r using CAPM, where r = R + β[E(R) – R]. Therefore, r = 3.5% + 1.6 × (9.0% F i M F – 3.5%) = 12.3% Step 3: calculate V = 3.0 / (12.3% – 5.25%) = $42.55. 0

C. $44.79. - Correct

Correct Answer Feedback: Correct because the intrinsic value of common stock is: V = D (1 + g)/(r – g) = D /(r – g) 0 0 1 Where: V = Value of a share of stock today 0 D = Expected dividend in year 1 1 r = Required rate of return on the stock g = Dividend growth rate. Step 1: calculate g, where: g = b × ROE, where b = earnings retention rate = (1 – dividend payout ratio). Therefore, b = (1 – 0.25) = 0.75. Therefore, g = 0.75 × 7% = 5.25%. Step 2: calculate D = D × (1 + g) = $3 × (1 + 5.25%) = $3.1575 1 0 Step 3: calculate r using CAPM, where r = R + β[E(R) – R], Therefore, r = 3.5% + 1.6 × F i M F (9.0% – 3.5%) = 12.3% Step 4: calculate V = 3.1575 / (12.3% – 5.25%) = $44.79. 0

Question 62

In general, which of the following institutions most likely has a high need for liquidity and a short investment time horizon?

Correct answer: A

A. Banks - Correct

Correct Answer Feedback: Correct because banks have a short-term horizon and high liquidity needs.

B. Endowments - Incorrect

Incorrect Answer Feedback: Incorrect because endowments have a long time horizon and low need for liquidity.

C. Defined-benefit pension plans - Incorrect

Incorrect Answer Feedback: Incorrect because defined-benefit pension plans have a long time horizon and low need for liquidity.

Question 63

The money duration of a bond is best defined as:

Correct answer: B

A. the bond's Macaulay duration times the bond's full price. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the annual modified duration, not Macaulay duration, times the bond's full price. Money duration (MoneyDur) is calculated as the annual modified duration times the full price (PVFull) of the bond.

B. the estimated change in the bond's full price in currency units for a given change in the annual yield to maturity. - Correct

Correct Answer Feedback: Correct because this describes the interpretation of money duration. The money duration of a bond is a measure of the price change in units of the currency in which the bond is denominated. For a given change in the annual yield-to- maturity (ΔYield), modified duration estimates the percentage price change and money duration estimates the change in currency units.

C. one half of the difference in the bond's full prices given a 1 basis point decrease and 1 basis point increase in yield-to-maturity. - Incorrect

Incorrect Answer Feedback: Incorrect because this describes the calculation of the price value of a basis point (PVBP). The PVBP is an estimate of the change in the full price given a 1 bp change in the yield-to-maturity. The PVBP can be calculated using a formula similar to that for the approximate modified duration, such that: PVBP = [(PV) − (PV)]/2 where PV− − + and PV+ are the full prices calculated by decreasing and increasing the yield-to-maturity by 1 bp.

Question 64

A special class of digital assets that peg their value to other financial assets through tokenization are called:

Correct answer: C

A. meme coins. - Incorrect

Incorrect Answer Feedback: Incorrect because meme coins are cryptocurrencies often inspired by a joke and are generally launched for entertainment purposes as they gain popularity in a short period of time, which allows early purchasers to sell their holdings at an often considerable profit.

B. security tokens. - Incorrect

Incorrect Answer Feedback: Incorrect because security tokens digitize the ownership rights associated with publicly traded securities. On the other hand, asset- backed token maintains price parity with some target asset, the US dollar or gold, through tokenization.

C. asset-backed tokens. - Correct

Correct Answer Feedback: Correct because a special class of digital assets is the asset-backed token, which maintains price parity with some target asset, the US dollar or gold, through tokenization.

Question 65

If the expected return on the market portfolio is 6% and the risk-free rate is 2%, the expected return of a security with a beta of 1.25 is closest to:

Correct answer: B

A. 5.00%. - Incorrect

Incorrect Answer Feedback: Incorrect. It fails to add the risk-free rate: 1.25 × (6% – 2%) = 5.00%.

B. 7.00%. - Correct

Correct Answer Feedback: Correct. The capital asset pricing model posits that the expected return of a security is E(R) = R + β[E(R) – R] where R is the risk-free rate, R is i f i m f f m the return on the market portfolio, and β is the beta of the security: 2% + 1.25 × (6% – 2%) = 7.00%.

C. 9.50%. - Incorrect

Incorrect Answer Feedback: Incorrect. It fails to subtract the risk-free rate from the market portfolio's expected return: 2% + 1.25 × (6%) = 9.50%.

Question 66

Compared to alternative investments, traditional investments are most likely characterized by:

Correct answer: C

A. greater use of leverage. - Incorrect

Incorrect Answer Feedback: Incorrect because alternative investments are more likely to have greater use of leverage. Investing in alternative investments is often pursued through such special vehicles as hedge funds and private equity funds, which have flexibility to use leverage.

B. a more complex fee structure. - Incorrect

Incorrect Answer Feedback: Incorrect because one way that alternative investments differ from traditional asset classes is that they involve more complex fee structures.

C. higher levels of transparency. - Correct

Correct Answer Feedback: Correct because alternative investments often have less regulation and less transparency than traditional investment.

Question 67

A 91-day US Treasury bill with a face value of $1,000,000 has a price of $985,000. Assuming a 360-day year, the discount rate is closest to:

Correct answer: B

A. 3.8%. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer transposes the Year and Days. The formula for the discount rate is DR = (Year/Days) × (FV–PV/FV). (91/360) × ((1,000,000–985,000)/1,000,000). 0.2528 × 0.015 = 0.0038 - incorrectly rounded as 3.8%.

B. 5.9%. - Correct

Correct Answer Feedback: Correct because the formula for the discount rate is DR = (Year/Days) × (FV–PV/FV). Solving the problem yields (360/91) × ((1,000,000– 985,000)/1,000,000). 3.956 × 0.015 = 0.0593 or 5.93%.

C. 6.0%. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer uses PV in the denominator rather than FV. The formula for the discount rate is DR = (Year/Days) × (FV– PV/FV). 360/91 × ((1,000,000 – 985,000) /985,000 = 3.956 × 0.01523 = 0.0602 ≈ 6.0%.

Question 68

Forward rate agreements can be used by firms to:

Correct answer: A

A. lock in a fixed borrowing or lending rate at a future date. - Correct

Correct Answer Feedback: Correct because forward contracts in which the underlying is an interest rate are called forward rate agreements, or FRAs. FRAs are forward contracts that allow participants to make a known interest payment at a later date and receive in return an unknown interest payment. In that way, a participant whose business will involve borrowing at a future date can hedge against an increase in interest rates by buying an FRA (the long side) and locking in a fixed payment and receiving a random payment that offsets the unknown interest payment it will make on its loan.

B. pay the return on an equity index and receive a fixed rate. - Incorrect

Incorrect Answer Feedback: Incorrect because paying the return on an equity index and receiving a fixed rate describes an equity swap, not a FRA. Equity swaps, sometimes called equity swaps, are quite popular and permit investors to pay the return on one stock index and receive the return on another index or a fixed rate.

C. hedge against exchange rate fluctuations in foreign-currency investments. - Incorrect

Incorrect Answer Feedback: Incorrect because FRAs are contracts in which the underlying is an interest rate and are used to lock in a fixed borrowing or lending rate at a future date. Therefore, FRAs hedge against interest rate risks, rather than currency fluctuations or exchange rate risks.

Question 69

Buying the stock of a company being acquired while going short the stock of the acquiring company best describes a(n):

Correct answer: B

A. activist strategy. - Incorrect

Incorrect Answer Feedback: Incorrect because the term “activist” is short for “activist shareholder.” Here, managers secure sufficient equity holdings to allow them to influence a company’s policies or direction. The hedge fund manager thus tries to create his or her own catalyst, influencing the investment’s ultimate destiny by creating a desired corporate outcome. For example, the activist hedge fund may advocate for divestitures, restructuring, capital distributions to shareholders, or changes in management and company strategy that will affect their equity holdings.

B. merger arbitrage strategy. - Correct

Correct Answer Feedback: Correct because a merger arbitrage strategy involves going long (buying) the stock of the company being acquired at a discount to its announced takeover price and going short (selling) the stock of the acquiring company when the merger or acquisition is announced.

C. distressed/restructuring strategy. - Incorrect

Incorrect Answer Feedback: Incorrect because a distressed/restructuring strategy focuses on the securities of companies that are either in bankruptcy or perceived to be near bankruptcy. One strategy sees hedge funds simply purchasing fixed-income securities that are trading at a significant discount to par but are still senior enough to be deemed “money good” (backed by enough corporate assets to be fully repayable at par or at least at a significant premium to the available bond purchase price) in a bankruptcy situation. Alternatively, a hedge fund may purchase the so-called fulcrum debt instrument that is expected to convert into new equity in the case of a restructuring or bankruptcy.

Question 70

Accrued interest:

Correct answer: B

A. depends on a bond’s yield-to-maturity. - Incorrect

Incorrect Answer Feedback: Incorrect because the accrued interest is part of the full price and does not depend on the yield-to-maturity.

B. is equal to a bond’s full price minus its flat price. - Correct

Correct Answer Feedback: Correct because PVFull = PVFlat + AI, where PVFull is the bond’s Flat full price, PV is the bond’s flat price, and AI is the accrued interest.

C. is typically included in the price quoted by bond dealers. - Incorrect

Incorrect Answer Feedback: Incorrect because the flat price usually is quoted by bond dealers. If a trade takes place, the accrued interest is added to the flat price to obtain the full price paid by the buyer and received by the seller on the settlement date.

Question 71

An investor sells a put option expiring in six months for $55.75. The put option has an exercise price of $190.50. Determine the profit for the investor if the price of the underlying at expiration is $195.00.

Correct answer: B

A. $0.00 - Incorrect

Incorrect Answer Feedback: Incorrect because it incorrectly assumes that the profit to the put seller is –Max(0, X – S) = –Max(0,$190.50 – $195.00) = –Max(0,–$4.50) = 0. As T described in the rationale for the correct answer, the profit to the put seller is Π = –Max(0,X – S) + p = T 0 –Max(0,$190.50 – $195.00) + $55.75 = –Max(0,–$4.50) + $55.75 = 0 + $55.75 = $55.75.

B. $55.75 - Correct

Correct Answer Feedback: Correct because the profit to the put seller is Π = –Max(0,X – S) T + p , where S is the price of the underlying at the expiration date, T, X is the exercise price of 0 T the option and p is the premium paid by the buyer to the seller. Therefore, the profit is 0 –Max(0,$190.50 – $195.00) + $55.75 = –Max(0,–$4.50) + $55.75 = 0 + $55.75 = $55.75.

C. $60.25 - Incorrect

Incorrect Answer Feedback: Incorrect because it incorrectly assumes that the profit to the put seller is Max(0, S – X) + p = Max(0,$195.00 – $190.50) + $55.75 = Max(0,$4.50) + T 0 $55.75 = $4.50 + $55.75 = $60.25. As described in the rationale for the correct answer, the profit to the put seller is Π = –Max(0,X – S) + p = –Max(0,$190.50 – $195.00) + $55.75 = T 0 –Max(0,–$4.50) + $55.75 = 0 + $55.75 = $55.75.

Question 72

An analyst gathers the following information about an equal-weighted index with three securities: Shares Beginning-of-Period End-of-Period Security Outstanding Price Price (Millions) 1 15 $5 $6 2 20 $2 $2 3 25 $3 $4 The price return of the index is closest to:

Correct answer: A

A. 17.8%. - Correct

Correct Answer Feedback: Correct because equal weighting assigns an equal weight to each constituent security at inception. To construct an equal-weighted index from three securities, the index provider allocates one third of the index value at the beginning of the period to each security. Assume the Index value at the beginning is 300 and 100 is allocated to each security, the equal-weighted return is calculated as follows: Shares in Shares EOP Value index in BOP BOP Weight EOP Security Price Value (%) Price (Shares × EOP (Value/BOP Index Price) Price) 1 1 5.00 100.00 33.33 20 6.00 120.00 2 1 2.00 100.00 33.33 50 2.00 100.00 3 1 3.00 100.00 33.33 33.33 4.00 133.33 Total (Index — — 300.00 100.00 — — 353.33 Value) Index Value — — 300 — — — 353.33 (Total/3) Therefore the equal-weighted price return is EOP index value / BOP index value – 1= 353.33/300 – 1≈ 0.1778 ≈ 17.8%.

Correct Answer Feedback: Correct because equal weighting assigns an equal weight to each constituent security at inception. To construct an equal-weighted index from three securities, the index provider allocates one third of the index value at the beginning of the period to each security. Assume the Index value at the beginning is 300 and 100 is allocated to each security, the equal-weighted return is calculated as follows:

Security	Shares in Index	BOP Price	BOP Value	Weight (%)	Shares in index (Value/BOP Price)	EOP Price	EOP Value (Shares × EOP Price)
1	1	5.00	100.00	33.33	20	6.00	120.00
2	1	2.00	100.00	33.33	50	2.00	100.00
3	1	3.00	100.00	33.33	33.33	4.00	133.33
Total (Index Value)	---	---	300.00	100.00	---	---	353.33
Index Value (Total/3)	---	---	300	---	---	---	353.33

Therefore the equal-weighted price return is EOP index value / BOP index value – 1= 353.33/300 – 1≈ 0.1778 ≈ 17.8%.

Figure 5: Source equation for question 72, option A

B. 20.0%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the price-weighted return calculated as follows: Shares in BOP Value EOP Value Security BOP Price EOP Price Index (Share × BOP Price) (Share × EOP Price) 1 1 5.00 5.00 6.00 6.00 2 1 2.00 2.00 2.00 2.00 3 1 3.00 3.00 4.00 4.00 Total — — 10.00 — 12.00 Index Value — — 3.33 — 4.00 (Total/3) The value of the price-weighted index is determined by dividing the sum of the security values by the divisor, which is typically set at inception to equal the initial number of securities in the index. Therefore the price-weighted price return = EOP index value / BOP index value – 1 = 4.00/3.33 – 1= 0.2 = 20%.

C. 21.1%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the market-capitalization weighted index. Market capitalization or value is calculated by multiplying the number of shares outstanding by the market price per share. BOP

Incorrect Answer Feedback: Incorrect because this is the price-weighted return calculated as follows:

Security	Shares in Index	BOP Price	BOP Value (Share × BOP Price)	EOP Price	EOP Value (Share × EOP Price)
1	1	5.00	5.00	6.00	6.00
2	1	2.00	2.00	2.00	2.00
3	1	3.00	3.00	4.00	4.00
Total	---	---	10.00	---	12.00
Index Value (Total/3)	---	---	3.33	---	4.00

The value of the price-weighted index is determined by dividing the sum of the security values by the divisor, which is typically set at inception to equal the initial number of securities in the index. Therefore the price-weighted price return = EOP index value / BOP index value – 1 = 4.00/3.33 – 1= 0.2 = 20%.

Figure 6: Source equation for question 72, option B

Value EOP Value Shares BOP Price EOP Price Security oustanding ($) (shares × ($) (shares × (Millions) BOP Price) EOP Price) 1 15 5.00 75 6.00 90 2 20 2.00 40 2.00 40 3 25 3.00 75 4.00 100 Total — — 190 — 230 Therefore the market-capitalization weighted price return is calculated as EOP index value / BOP index value – 1= 230/190 – 1 = 0.2105 ≈ 21.1%.

Question 73

Which of the following statements about open-end mutual funds and ETFs is most accurate?

Correct answer: C

A. Investors can buy both open-end mutual funds and ETFs from other investors - Incorrect

Incorrect Answer Feedback: Incorrect because mutual fund investors buy the fund shares directly from the fund. An alternative to setting the fund up as an open-end fund would be to create a closed-end fund in which no new investment money is accepted into the fund. New investors invest by buying existing shares, and investors in the fund liquidate by selling their shares to other investors.

B. Open-end mutual funds can be priced intraday, while ETFs are priced once a day - Incorrect

Incorrect Answer Feedback: Incorrect because ETFs can be transacted (and are priced) intraday. In contrast, mutual funds typically can be purchased or sold only once a day.

C. ETFs can be settled at a price different from the NAV, while open-end mutual funds are settled at the NAV - Correct

Correct Answer Feedback: Correct because mutual fund investors buy the fund shares directly from the fund, and all investments are settled at the net asset value. In practice, the market price of the ETF is likely to be close (but different) to the net asset value of the underlying investments. The mutual fund can be set up as an open-end fund or a closed-end fund. If it is an open-end fund, it will accept new investment money and issue additional shares at a value equal to the net asset value of the fund at the time of investment.

Question 74

Which of the following contracts is an example of a contingent claim? A(n):

Correct answer: A

A. currency option - Correct

Correct Answer Feedback: Correct because there are two general classes of derivatives. Some provide the ability to lock in a price at which one might buy or sell the underlying. Because they force the two parties to transact in the future at a previously agreed-on price, these instruments are called forward commitments. The various types of forward commitments are called forward contracts, futures contracts, and swaps. Another class of derivatives provides the right but not the obligation to buy or sell the underlying at a pre- determined price. Because the choice of buying or selling versus doing nothing depends on a particular random outcome, these derivatives are called contingent claims. The primary contingent claim is called an option. Therefore, a currency option is an example of a contingent claim.

B. interest rate swap - Incorrect

Incorrect Answer Feedback: Incorrect because swaps are forward commitments and not contingent claims.

C. commodity forward - Incorrect

Incorrect Answer Feedback: Incorrect because forward contracts are forward commitments and not contingent claims.

Question 75

The Markowitz efficient frontier is best described as the part of the minimum-variance frontier of risky assets that plots:

Correct answer: B

A. to the right of the risk-free asset. - Incorrect

Incorrect Answer Feedback: Incorrect because the minimum-variance frontier gives us portfolios with the minimum variance for a given return. The curve that lies above and to the right of the global minimum-variance portfolio (and not the risk-free asset) is referred to as the Markowitz efficient frontier because it contains all portfolios of risky assets that rational, risk-averse investors will choose. In fact, all portfolios of the minimum-variance frontier lie to the right of the risk-free asset.

B. above the global minimum-variance portfolio. - Correct

Correct Answer Feedback: Correct because the minimum-variance frontier gives us portfolios with the minimum variance for a given return. The curve that lies above and to the right of the global minimum-variance portfolio is referred to as the Markowitz efficient frontier because it contains all portfolios of risky assets that rational, risk-averse investors will choose.

C. above and to the right of the optimal risky portfolio. - Incorrect

Incorrect Answer Feedback: Incorrect because the minimum-variance frontier gives us portfolios with the minimum variance for a given return. The curve that lies above and to the right of the global minimum-variance portfolio [and not the optimal risky portfolio] is referred to as the Markowitz efficient frontier because it contains all portfolios of risky assets that rational, risk-averse investors will choose. The optimal risky portfolio is at the tangent of the capital allocation line and the efficient frontier of risky assets, and thus is not the same as the global minimum-variance portfolio.

Question 76

Which of the following is most likely an example of a Eurobond?

Correct answer: C

A. A Japanese borrower issuing US dollar–denominated bonds in the US market. - Incorrect

Incorrect Answer Feedback: Incorrect because this is an example of a foreign bond, which is a bond issued by an entity incorporated in a country other than where the bond is issued and in whose currency the bond is denominated.

B. A Canadian borrower issuing British pound–denominated bonds in the UK market. - Incorrect

Incorrect Answer Feedback: Incorrect because this is an example of a foreign bond, which is a bond issued by an entity incorporated in a country other than where the bond is issued and in whose currency the bond is denominated.

C. An Australian borrower issuing Canadian dollar–denominated bonds in the UK market. - Correct

Correct Answer Feedback: Correct. A Eurobond is an international bond issued outside the jurisdiction of any one country and not denominated in the currency of the country where it is issued.

Question 77

Which of the following is the best example of an embedded option granted to bondholders?

Correct answer: B

A. A prepayment option - Incorrect

Incorrect Answer Feedback: Incorrect because this is an embedded option granted to the issuer who can call a portion of the issue.

B. A put if the issuer's rating changes - Correct

Correct Answer Feedback: Correct because a put allows the bondholder to sell the security back to the issuer if the bondholder chooses to do so.

C. An increasing sinking fund provision - Incorrect

Incorrect Answer Feedback: Incorrect because this is an embedded option granted to the issuer who can call more than is necessary to meet the sinking fund requirement.

Question 78

In derivatives pricing:

Correct answer: B

A. investors are assumed to be risk averse. - Incorrect

Incorrect Answer Feedback: Incorrect because the important point to understand is that while the risk aversion of investors is relevant to pricing assets, it is not relevant to pricing derivatives. As such, derivatives pricing is sometimes called risk-neutral pricing. Therefore, investors are not assumed to be risk averse.

B. a portfolio consisting of the underlying and the derivative must earn the risk- free rate. - Correct

Correct Answer Feedback: Correct because the derivative price is the price that guarantees the risk-free combination of the derivative and the underlying produces a risk- free return. Also, risk-neutral pricing uses the fact that arbitrage opportunities guarantee that a risk-free portfolio consisting of the underlying and the derivative must earn the risk- free rate.

C. expected payoffs of the derivative are discounted at the risk-free rate plus a risk premium. - Incorrect

Incorrect Answer Feedback: Incorrect. The risk aversion of the investor is not relevant to pricing the derivative; therefore, one can just as easily obtain the derivative price by assuming that the investor is risk neutral. That means that the expected payoff of the derivative can be discounted at the risk-free rate rather than the risk-free rate plus a risk premium.

Question 79

A value-weighted equity index is most likely biased by a stock that:

Correct answer: C

A. has a high price. - Incorrect

Incorrect Answer Feedback: Incorrect because neither stock prices nor splits have an impact on a value-weighted index.

B. splits frequently. - Incorrect

Incorrect Answer Feedback: Incorrect because neither stock prices nor splits have an impact on a value-weighted index.

C. has a large market capitalization. - Correct

Correct Answer Feedback: Correct because the impact of a given percentage price change of a large-cap stock is larger than the impact of the same percentage price change of a small-cap stock, which is a source of bias for value-weighted indexes.

Question 80

Which of the following risk metrics reflects the risk of changes in an option's delta?

Correct answer: C

A. Rho - Incorrect

Incorrect Answer Feedback: Incorrect because rho measures the change in the option price for a change in interest rates, not another risk metric. Derivatives are also sensitive to changes in interest rates, which are reflected in a measure called rho.

B. Vega - Incorrect

Incorrect Answer Feedback: Incorrect because vega measures the change in the option price for a change in volatility, not another risk metric. Some derivatives, such as options, are also sensitive to changes in the volatility of the underlying. This risk is captured by a concept called vega, which is a first-order measure of the change in the derivative price for a change in the volatility of the underlying.

C. Gamma - Correct

Correct Answer Feedback: Correct because the sensitivity of the derivative price to a small change in the value of the underlying asset is called the delta. It is perhaps the most important measure of derivatives risk. Yet delta is limited to capturing only small changes in the value of the underlying. Large changes are captured by the concept of gamma. Whereas delta is a first-order risk, gamma is considered a second-order risk because it reflects the risk of changes in delta.

Question 81

Before performance fees can be charged on newly generated profits when a hedge fund experience losses, the value of the hedge fund must exceed the:

Correct answer: C

A. hurdle rate. - Incorrect

Incorrect Answer Feedback: Incorrect because the partnership agreement usually specifies that the performance fee is earned only after the fund achieves a return known as a hurdle rate.

B. carried interest. - Incorrect

Incorrect Answer Feedback: Incorrect because a performance fee (also referred to as an incentive fee or carried interest) is applied based on excess returns.

C. high-water mark. - Correct

Correct Answer Feedback: Correct because in hedge funds, fee calculations also take into account a high-water mark, which reflects the highest value used to calculate an incentive fee. A high-water mark is the highest value of the fund investment ever achieved at a performance fee crystallization date, net of fees, by the individual LP. A high-water mark clause states that a hedge fund manager must recuperate declines in value from the high- water mark before performance fees can be changed on newly generated profits.

Question 82

An analyst gathers the following information about a hedge fund's first year of operations:

Beginning-of-year assets under management	$100,000,000
Management fee based on end-of-year assets under management	2%
Incentive fee, calculated net of the management fee	20%
Annual gross return	15%
Soft hurdle rate	5%

The total fees (in $ millions) earned by the hedge fund in Year 1 are:

Correct answer: C

A. 3.84. - Incorrect

Incorrect Answer Feedback: Incorrect because the soft hurdle rate is treated as a hard hurdle rate. The management fee is $2,300,000 ($115,000,000× .02) and the incentive fee is $1,540,000 (($115,000,000 – $100,000,000 – $2,300,000 – $5,000,000 hurdle rate) × .2). Total fees are $3,840,000.

B. 4.30. - Incorrect

Incorrect Answer Feedback: Incorrect because the hurdle rate is treated as a hard hurdle rate and the incentive fee is calculated independently of the management fee. The management fee is $2,300,000 ($115,000,000 × .02) and the incentive fee is $2,000,000 (($115,000,000 – $100,000,000 – $5,000,000 hurdle rate) × .2). Total fees are $4,300,000.

C. 4.84. - Correct

Correct Answer Feedback: Correct because the management fee is $2,300,000 ($115,000,000 × .02) and the incentive fee is $2,540,000 (($115,000,000 – $100,000,000 – $2,300,000) × .2). Total fees are $4,840,000. With a soft hurdle rate, the incentive fee is earned on all of the return as long as the hurdle rate is exceeded.

Question 83

The price of an option-free bond increases by 5% when the yield to maturity decreases by 200 basis points. If the price of this bond decreases by 5%, the yield to maturity most likely increases by:

Correct answer: C

A. less than 200 basis points. - Incorrect

Incorrect Answer Feedback: Incorrect because for the same coupon rate and time-to- maturity, the percentage price change is greater (in absolute value, meaning without regard to the sign of the change) when the market discount rate goes down than when it goes up (the convexity effect).

B. 200 basis points. - Incorrect

Incorrect Answer Feedback: Incorrect because for the same coupon rate and time-to- maturity, the percentage price change is greater (in absolute value, meaning without regard to the sign of the change) when the market discount rate goes down than when it goes up (the convexity effect).

C. more than 200 basis points. - Correct

Correct Answer Feedback: Correct because for an option-free bond with the same coupon rate and time-to-maturity, the percentage price change is greater (in absolute value, meaning without regard to the sign of the change) when the market discount rate goes down than when it goes up (the convexity effect).

Question 84

A company increased the dividend paid on its common stock by 3% per year over the past three years. If this growth rate is expected to continue indefinitely, the most appropriate model to estimate the intrinsic value of the stock is the:

Correct answer: A

A. Gordon growth model. - Correct

Correct Answer Feedback: Correct because the Gordon growth model assumes perpetual constant growth. Because of its assumption of a constant growth rate, the Gordon growth model is particularly appropriate for valuing the equity of dividend-paying companies that are relatively insensitive to the business cycle and in a mature growth phase. A history of increasing the dividend at a stable growth rate is another practical criterion if the analyst believes that pattern will hold in the future.

B. two-stage dividend discount model. - Incorrect

Incorrect Answer Feedback: Incorrect because a two-stage model assumes that at some point the company will begin to pay dividends that grow at a constant rate, but prior to that time the company will pay dividends that are growing at a higher rate than can be sustained in the long run. Since the firm has been paying dividends that have been increasing and the pattern is expected to continue, the Gordon growth model is more suitable since it assumes that the dividend growth rate is forever: It is perpetual and never changes.

C. three-stage dividend discount model. - Incorrect

Incorrect Answer Feedback: Incorrect because a three-stage DDM would be most appropriate for a fairly young company, one that is just entering the growth phase. Since the firm operates in an industry that is relatively insensitive to the business cycle and has been paying a steady stream of increasing dividends, the Gordon growth model is more suitable since it assumes that the dividend growth rate is forever: It is perpetual and never changes.

Question 85

Credit card receivable asset-backed securities (ABS) are most likely to:

Correct answer: A

A. be secured by non-amortizing loans. - Correct

Correct Answer Feedback: Correct because the collateral of credit card receivable ABS is a pool of non-amortizing loans. These loans have lockout periods during which the cash flows that are paid out to security holders are based only on finance charges collected and fees. When the lockout period is over, the principal that is repaid by the cardholders is no longer reinvested but instead is distributed to investors.

B. have principal repayments reinvested indefinitely. - Incorrect

Incorrect Answer Feedback: Incorrect because when the lockout period is over, the principal that is repaid by the cardholders is no longer reinvested but instead is distributed to investors.

C. expose investors to contraction risk during a lockout period. - Incorrect

Incorrect Answer Feedback: Incorrect because the collateral of credit card receivable ABS is a pool of non-amortizing loans. These loans have lockout periods during which the cash flows that are paid out to security holders are based only on finance charges collected and fees. When the lockout period is over, the principal that is repaid by the cardholders is no longer reinvested but instead is distributed to investors. During the lock-out period, investors are not exposed to contraction risk as principal repayments are reinvested, not distributed.

Question 86

Which of the following alternative investment categories most likely provides exposure to capital-intensive, long-lived assets that provide essential public services?

Correct answer: A

A. Infrastructure - Correct

Correct Answer Feedback: Correct because infrastructure is a special type of real asset that typically involves land, buildings and other long-lived fixed assets that are intended for public use and provide essential services. Also, infrastructure consists of assets that are capital intensive and long lived and that are intended to provide essential services for public use.

B. Venture capital - Incorrect

Incorrect Answer Feedback: Incorrect because venture capital is a specialized form of private equity whereby ownership capital is used for non-public companies in the early life cycle or startup phase, where often an idea or business plan exists with a limited operation or customer base.

C. Commercial real estate - Incorrect

Incorrect Answer Feedback: Incorrect because commercial real estate includes land and buildings where private business activity is the primary cash flow source. Although commercial real estate assets are long-lived, real assets, they typically do not provide essential services.

Question 87

In contrast to investors in otherwise-similar mortgage-backed securities, investors in covered bonds most likely:

Correct answer: C

A. face higher credit risks. - Incorrect

Incorrect Answer Feedback: Incorrect because in contrast to a static pool of mortgage loans that expose investors to prepayment risk as in the case of US mortgage-backed securities, cover pool sponsors must replace any prepaid or non-performing assets (i.e., assets that do not generate the promised cash flows) in the cover pool to ensure sufficient cash flows until the maturity of the covered bond.

B. are exposed to prepayment risks. - Incorrect

Incorrect Answer Feedback: Incorrect because in contrast to a static pool of mortgage loans that expose investors to prepayment risk as in the case of US mortgage-backed securities, cover pool sponsors must replace any prepaid or non-performing assets (i.e., assets that do not generate the promised cash flows) in the cover pool to ensure sufficient cash flows until the maturity of the covered bond.

C. have recourse against the issuing financial institution. - Correct

Correct Answer Feedback: Correct because covered bonds are similar to ABS but offer bondholders dual recourse—that is, to both the issuing financial institution and the underlying asset pool.

Question 88

A behavioral bias in which an investor assesses probabilities of outcomes depending on how similar they are to the current state is called:

Correct answer: C

A. conservatism. - Incorrect

Incorrect Answer Feedback: Incorrect because conservatism is where investors tend to be slow to react to changes.

B. narrow framing. - Incorrect

Incorrect Answer Feedback: Incorrect because in narrow framing, investors focus on issues in isolation.

C. representativeness. - Correct

Correct Answer Feedback: Correct because an investor assessing probabilities of outcomes depending on how similar they are to the current state is referred to as representativeness.

Question 89

When valuing underlying hedge fund positions, the most conservative approach most likely uses:

Correct answer: C

A. the most recent market prices. - Incorrect

Incorrect Answer Feedback: Incorrect because when market prices or quotes are used for valuation, funds may differ in which price or quote they use (for example, bid price, ask price, average quote, and median quote).

B. the average of the bid and ask prices. - Incorrect

Incorrect Answer Feedback: Incorrect because although using the average quote [(bid + ask)/2] is a common approach, a more conservative and theoretically accurate approach is to use bid prices for longs and ask prices for shorts as these are the prices at which the positions could be closed.

C. bid prices for longs and ask prices for shorts. - Correct

Correct Answer Feedback: Correct because a conservative and theoretically accurate approach is to use bid prices for longs and ask prices for shorts because these are the prices at which the positions could be closed.

Question 90

An investor makes the following statements: • Statement 1: “Swaps are contingent claims.” • Statement 2: “Swaps are characterized by a series of cash flows.” Which of the statements is correct?

Correct answer: B

A. Statement 1 only - Incorrect

Incorrect Answer Feedback: Incorrect because the various types of forward commitments are called forward contracts, futures contracts, and swaps. Therefore, swaps are commitments, not contingent claims.

B. Statement 2 only - Correct

Correct Answer Feedback: Correct because a swap is an over-the-counter derivative contract in which two parties agree to exchange a series of cash flows whereby one party pays a variable series that will be determined by an underlying asset or rate and the other party pays either (1) a variable series determined by a different underlying asset or rate or (2) a fixed series.

C. Both Statement 1 and Statement 2 - Incorrect

Incorrect Answer Feedback: Incorrect because the various types of forward commitments are called forward contracts, futures contracts, and swaps. Therefore, swaps are commitments, not contingent claims.